CBSE CLASS 12
BIOLOGY

Comprehensive Answer Booklet

Most Expected Questions — Board Exam 2026

Chapters Covered:

Ch 2 · Ch 3 · Ch 4 · Ch 5 · Ch 6 · Ch 7
Ch 8 · Ch 10 · Ch 11 · Ch 12 · Ch 13 · Ch 14 · Ch 15

Sexual Reproduction · Human Reproduction · Reproductive Health
Genetics & Variation · Molecular Biology · Evolution
Health & Disease · Microbes · Biotechnology
Ecology · Biodiversity & Conservation

TOPPERS 24/7 | PM Shri KV Khairagarh | 2025–26

Chapter 2 — Sexual Reproduction in Flowering Plants

Q1

Angiosperms bearing unisexual flowers are said to be either monoecious or dioecious. Explain with the help of one example each.

Answer:

Monoecious Plants: Bear both male (staminate) and female (pistillate) unisexual flowers on the **same plant**.

- Example: **Maize (*Zea mays*)** — tassels (male flowers) at the top; cobs (female flowers) at lower nodes.
- Other examples: *Cucurbita* (pumpkin/cucumber), *Colocasia*.

Dioecious Plants: Bear male and female flowers on **different (separate) plants**.

- Example: **Papaya (*Carica papaya*)** — some plants bear only male flowers; others only female.
- Other example: Date palm (*Phoenix dactylifera*).

⋆ Key Point

Monoecious = both sexes on *same* plant — **Dioecious** = sexes on *different* plants

Q2

Draw a diagram of a mature microspore of an angiosperm. Label its cellular components only. *OR* Draw a labelled diagram of a mature pollen grain.

Answer:

Structure of Mature Pollen Grain (2-celled stage):

- **Exine:** Outer hard layer made of sporopollenin; has germ pores.
- **Intine:** Inner thin layer made of cellulose and pectin.
- **Vegetative cell:** Large cell with lobed/irregular nucleus and abundant food reserves.
- **Generative cell:** Small, spindle-shaped cell that floats in the cytoplasm of the vegetative cell (like a cell within a cell).

Diagram labels: → Exine → Intine → Germ pore → Vegetative cell (large nucleus) → Generative cell (small, spindle)

⋆ Key Point

Sporopollenin in the exine is one of the most resistant biological substances known — pollen is excellently preserved in fossil records (Palynology).

Q3

A bilobed, dithecous anther has 100 microspore mother cells per microsporangium. How many male gametophytes can this anther produce?

Answer:

- Bilobed anther → **2 lobes**
- Dithecous → **2 thecae per lobe** → Total **4 microsporangia**
- Each microsporangium has **100 Microspore Mother Cells (MMC)**
- Each MMC undergoes **Meiosis → 4 haploid microspores**
- Each microspore → **1 male gametophyte (pollen grain)**

$$\boxed{\text{Total male gametophytes} = 4 \times 100 \times 4 = \mathbf{1600}}$$

⋆ **Key Point**

Bilobed (2 lobes) × Dithecous (2 thecae/lobe) = 4 microsporangia. Each MMC $\xrightarrow{\text{Meiosis}}$ 4 microspores → 4 pollen grains.

Q4

A pollen grain could be 2-celled or 3-celled at dehiscence. Explain how cells are placed within the pollen grain when shed at the 2-celled stage.

Answer:

When shed at the **2-celled stage**, the pollen grain contains:

- One large **vegetative cell (tube cell):** Large, with abundant food reserves and an irregular lobed nucleus.
- One small **generative cell:** Small, spindle-shaped, with dense cytoplasm — **embedded within the cytoplasm of the vegetative cell** (cell within a cell arrangement).

In the **3-celled stage**, the generative cell has already divided by mitosis into **two male gametes** before shedding.

⋆ **Key Point**

Most angiosperms shed pollen at the 2-celled stage. The generative cell divides later inside the pollen tube during germination.

Q5

Describe the process of megasporogenesis up to fully developed embryo sac formation in an angiosperm. (All India 2019)

Answer:

Megasporogenesis:

1. A single hypodermal cell in the **nucellus** of the ovule differentiates into the **Megaspore Mother Cell (MMC)** — large, with dense cytoplasm and a prominent nucleus.
2. MMC undergoes **Meiosis I + Meiosis II** → linear tetrad of **4 haploid megaspores**.
3. Three megaspores **degenerate**. The megaspore at the **chalazal end** remains functional.

Embryo Sac Formation (Female Gametophyte):

1. The functional megaspore undergoes **3 successive mitotic divisions**:
   - 1st mitosis → 2 nuclei (one at each pole)
   - 2nd mitosis → 4 nuclei (2 at each pole)
   - 3rd mitosis → 8 nuclei (4 at each pole)
2. **Organisation of 8 nuclei into 7 cells:**
   - *Micropylar end:* **Egg apparatus** = 1 egg cell + 2 synergids (3 cells)
   - *Chalazal end:* **3 antipodal cells**
   - *Centre:* 2 polar nuclei fuse → **Secondary nucleus** (1 central cell)

⋆ **Key Point**

Final embryo sac = **7 cells, 8 nuclei** (Polygonum/monosporic type). Process: MMC $\xrightarrow{\text{Meiosis}}$ 4 megaspores $\xrightarrow{\text{3 mitoses}}$ 8-nucleate embryo sac.

Q6

Describe the sequence of microsporogenesis in angiosperms. Draw a labelled diagram of the 2-celled final structure.

Answer:

1. In the young anther, **sporogenous tissue** differentiates in each microsporangium.
2. Each sporogenous cell is a potential **Microspore Mother Cell (Pollen Mother Cell, PMC)**.
3. PMC undergoes **Meiosis → tetrad of 4 haploid microspores**.
4. Microspores separate from tetrad; each develops thick wall:
   - Outer **exine** (sporopollenin) with germ pores
   - Inner **intine** (cellulose + pectin)
5. Nucleus divides by **Mitosis I** → large vegetative cell + small generative cell = **2-celled pollen grain**.

Diagram labels (2-celled pollen grain): Exine | Intine | Germ pore | Vegetative cell (irregular nucleus) | Generative cell (spindle-shaped, inside vegetative cell)

⋆ **Key Point**

Sporogenous tissue $\xrightarrow{\text{Meiosis}}$ Microspore tetrad $\xrightarrow{\text{Mitosis}}$ 2-celled pollen grain. Sporopollenin resists strong acids and high temperatures.

Q7

List the different types of pollination depending upon the source of pollen grain. Differentiate between autogamy, geitonogamy and xenogamy.

Answer:

Types of Pollination:

- **Autogamy** — pollen from the *same flower*

- **Geitonogamy** — pollen from a *different flower of the same plant*
- **Xenogamy** — pollen from a flower of a *genetically different plant*

Feature	Autogamy	Geitonogamy	Xenogamy
Source of pollen	Same flower	Different flower, same plant	Different plant (same species)
Genetic variation	No	No (genetically self)	Yes (true variation)
Pollinating agent	Not needed	Usually needed	Always needed
Significance	Ensures seed set	Functionally cross but genetically self	True cross-pollination

⋆ Key Point

Xenogamy is the *only* type that leads to true genetic recombination and new combinations of alleles.

Q8

Draw an LS of the pistil showing the pollen tube entering the embryo sac. Label Nucellus, Antipodals, Synergids, Micropyle. Write functions of Synergids and Micropyle.

Answer:

Diagram labels (LS of pistil): Stigma → Style → Ovary wall → Funiculus → Micropyle → Nucellus → Pollen tube → Synergids → Egg cell → Polar nuclei (Central cell) → Antipodals → Chalaza

Functions:

- **(a) Synergids:**
  - Possess special cellular thickenings called **filiform apparatus** at the micropylar end.
  - Act as chemotropic guide for pollen tube navigation into the embryo sac.
  - Degenerate after fertilisation.
- **(b) Micropyle:**
  - A small pore at the micropylar end of the ovule through which **pollen tube enters** the embryo sac.
  - Facilitates absorption of **water and oxygen** during seed germination.

⋆ Key Point

Filiform apparatus in synergids secretes chemicals that guide the pollen tube — chemotropism in fertilisation.

Q9

(i) What are false fruits? Give an example. (ii) What is parthenocarpy? Name a fruit developed by this method. (iii) Values shown by Ashu?

Answer:

- **(i) False Fruits:** Fruits in which parts *other than the ovary* (e.g., thalamus/receptacle) contribute to fruit formation.
  Example: **Apple** (*Malus domestica*) — the edible fleshy part develops from the **thalamus**, not the ovary. Other examples: Strawberry, Pear, Cashew.
- **(ii) Parthenocarpy:** Development of fruit *without fertilisation*. These fruits are **seedless**. Can be induced by growth hormones (gibberellins, auxins).
  Example: **Banana** (natural parthenocarpy); seedless grapes (induced).
- **(iii) Values shown by Ashu:**
  - Scientific curiosity and inquisitiveness
  - Willingness to learn beyond textbooks
  - Seeking knowledge from teachers (respect for guidance)

Chapter 3 — Human Reproduction

Q1

Why are human testes located outside the abdominal cavity? Name the pouch in which they are present.

Answer:

Human testes are located *outside* the abdominal cavity because **spermatogenesis requires a temperature 2–2.5°C lower** than normal body temperature (37°C). The abdominal temperature is too high for sperm production.
The pouch in which the testes are located is called the **Scrotum**.

⋆ Key Point

Cryptorchidism = failure of testes to descend into scrotum → leads to male infertility (sperm cannot develop at high temperature).

Q2

List the different parts of the human oviduct through which the ovum travels till it meets the sperm for fertilisation.

Answer:

After ovulation, the ovum travels through the following parts:

1. **Infundibulum** — funnel-shaped opening with finger-like **fimbriae** that collect the ovum from the ovary surface.
2. **Ampulla** — widest and longest part; **site of fertilisation**.
3. **Isthmus** — narrow part connecting to the uterus.
4. **Uterus (Womb)** — site of **implantation** of the blastocyst.

⋆ Key Point

Fertilisation = Ampulla of Fallopian tube | Implantation = Uterine wall (endometrium)

Q3

Differentiate between vas deferens and vasa efferentia.

Answer:

Poi	Vas Deferens	Vasa Efferentia
Ori	Arises from epididymis	Arise from rete testis
Fu	Carries sperm from epididymis to ejaculatory duct	Carry sperm from rete testis to epididymis
Nu	Single tube per testis	Multiple (10–15) small tubules
Le	Longer duct	Short, coiled tubules
Loc	Part of spermatic cord	Remain within testicular region

Q4

Explain the role of pituitary and sex hormones in the process of spermatogenesis. (Hormonal Regulation)

Answer:

1. **GnRH** (Gonadotropin Releasing Hormone) from **hypothalamus** → stimulates anterior pituitary.
2. **FSH** (Follicle Stimulating Hormone) from anterior pituitary → acts on **Sertoli cells** → supports and nourishes developing sperm.
3. **LH** (Luteinizing Hormone / ICSH in males) from anterior pituitary → acts on **Leydig cells (interstitial cells)** → stimulates secretion of **testosterone**.
4. **Testosterone** (androgen) → essential for **initiating and maintaining spermatogenesis**; also responsible for secondary sexual characters.

⋆ Key Point

Hypothalamus $\xrightarrow{\text{GnRH}}$ Pituitary $\xrightarrow{\text{FSH + LH}}$ Sertoli cells + Leydig cells → Spermatogenesis + Testosterone

Q5

Draw and label the parts of the head region only of a human sperm.

Answer:

Head region of human sperm contains:

- **Acrosome:** Cap-like structure at the very tip of the head, derived from Golgi body. Contains **hydrolytic enzymes** (hyaluronidase, acrosin) that help penetrate the zona pellucida of the egg during fertilisation.
- **Nucleus:** Large, flat, oval structure occupying most of the head. Contains **haploid (n) DNA** — the paternal genetic material.
- **Plasma membrane:** Encloses the entire head region.

Diagram labels: Acrosome (tip) → Nucleus (large oval) → Plasma membrane

⋆ Key Point

Acrosome reaction: On contact with zona pellucida, acrosomal enzymes are released (exocytosis) to enable sperm penetration.

Q6

Differentiate between menarche and menopause.

Answer:

Menarche	Menopause
First menstrual cycle in a female's life	Permanent cessation of the menstrual cycle
Marks the *beginning* of reproductive period	Marks the *end* of reproductive period
Occurs around age **10–13 years**	Occurs around age **45–50 years**
Onset of puberty in females	Associated with declining ovarian function

Q7

Explain the events in a normal woman during her menstrual cycle on (i) Days 13–15 (ii) Days 16–23 (iii) Days 24–29.

Answer:

- **(i) Ovarian events from Days 13–15:**
  - **LH surge** on day 13–14 triggers **ovulation** on day 14.
  - The Graafian follicle ruptures; secondary oocyte is released into the oviduct.
- **(ii) Ovarian hormones from Days 16–23 (Luteal/Secretory phase):**
  - Ruptured follicle transforms into **corpus luteum**.
  - Corpus luteum secretes large amounts of **progesterone** and some **oestrogen**.
  - These hormones maintain and thicken the endometrium for possible implantation.
- **(iii) Uterine events from Days 24–29:**
  - If fertilisation does not occur → corpus luteum degenerates.
  - Progesterone and oestrogen levels **drop sharply**.
  - Endometrium breaks down → **menstruation begins** (days 28–29).

⋆ Key Point

Day 1–5: Menstrual phase | Day 6–13: Follicular/Proliferative | Day 14: Ovulation | Day 15–28: Luteal/Secretory

Q9

Explain the process of spermatogenesis in humans. Draw a human sperm and label the acrosome and middle piece. Mention their functions.

Answer:

Spermatogenesis:

1. **Spermatogonia** (2n) — diploid stem cells in seminiferous tubules.
2. Spermatogonia undergo **Mitosis** → some become **Primary Spermatocytes** (2n).
3. Primary spermatocytes undergo **Meiosis I** → 2 **Secondary Spermatocytes** (n).
4. Secondary spermatocytes undergo **Meiosis II** → 4 **Spermatids** (n).
5. **Spermiogenesis:** Spermatids transform into mature **spermatozoa** (elongation, loss of cytoplasm, tail formation).
6. **Spermiation:** Mature sperm released from Sertoli cells into seminiferous tubule lumen.

Functions:

- **Acrosome:** Contains hydrolytic enzymes (hyaluronidase, acrosin) → helps penetrate zona pellucida of egg during fertilisation.
- **Middle piece:** Contains numerous **mitochondria** arranged helically → provides **energy (ATP)** for sperm motility (flagellar movement).

⋆ **Key Point**

1 Primary spermatocyte $\xrightarrow{\text{Meiosis I + II}}$ 4 Spermatids $\xrightarrow{\text{Spermiogenesis}}$ 4 Spermatozoa

Q10

Briefly explain the events of fertilisation and implantation. Comment on the role of the placenta as an endocrine gland.

Answer:

Fertilisation:

- Occurs in the **ampulla of the fallopian tube**.
- Sperm reaches the ovum; acrosome reaction releases enzymes → penetration of zona pellucida.
- Sperm nucleus fuses with egg nucleus → **zygote** (2n) formed. This triggers completion of Meiosis II in secondary oocyte.

Implantation:

- Zygote undergoes mitotic cleavage → **morula** → **blastocyst**.
- Blastocyst reaches uterus (day 5–6), loses zona pellucida (hatching).
- Inner cell mass (embryoblast) attaches to **endometrium** → implantation (day 7).

Placenta as Endocrine Gland:

Placenta secretes:

- **hCG** (human Chorionic Gonadotropin) — maintains corpus luteum in early pregnancy.
- **Progesterone** — maintains pregnancy, prevents uterine contractions.
- **Oestrogen** — promotes foetal growth.
- **hPL** (human Placental Lactogen) — metabolic adjustments during pregnancy.

⋆ Key Point

hCG is detected in urine → basis of **pregnancy test kits**. Placenta = endocrine + nutritive + excretory organ.

Chapter 4 — Reproductive Health

Q1

Our government has intentionally imposed strict conditions for MTP in our country. Justify giving a reason.

Answer:

MTP (Medical Termination of Pregnancy) has strict legal conditions because:

- To prevent **female foeticide** and sex-selective abortions (misuse of the technology).
- India has historically faced a skewed sex ratio; MTP was being misused to abort female foetuses.
- To protect the health and life of the mother from unsafe/illegal abortions.
- MTP is legal only up to **20 weeks** of pregnancy (extendable to 24 weeks in special cases) under the **Medical Termination of Pregnancy Act**.
- Must be performed only by **registered medical practitioners** at approved centres.

⋆ Key Point

MTP Act + PCPNDT Act together protect women's health and prevent sex determination/foeticide.

Q2

Describe the lactational amenorrhea method of birth control.

Answer:

- **Lactational amenorrhea** = absence of menstruation during intense breastfeeding after childbirth.
- During lactation, high **prolactin** levels → inhibit GnRH → suppress LH and FSH → **no ovulation**.
- Provides natural contraception for up to **6 months after childbirth**, BUT only if:
  - The mother is **exclusively breastfeeding** (no supplementary food to baby)
  - No menstrual cycle has resumed
- Natural, hormone-free, cost-free method with no side effects.
- Effectiveness *decreases* after 6 months or once supplementary feeding begins.

Q4

Why is there a statutory ban on amniocentesis? Write its use and justify the ban.

Answer:

Uses of Amniocentesis:

- Amniotic fluid (containing foetal cells) is collected and analysed chromosomally and

biochemically.

- Detects **chromosomal disorders**: Down's syndrome (trisomy 21), Turner's syndrome (45, XO), Klinefelter's syndrome (47, XXY).
- Detects **metabolic/biochemical disorders** and sex-linked genetic diseases.
- Can determine the sex of the foetus.

Reason for Statutory Ban:

- Amniocentesis was being **misused for sex determination** → leading to **female foeticide**.
- This severely worsened India's sex ratio.
- Hence, its use for sex determination is banned under the **PCPNDT Act** (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994).

⋆ Key Point

Amniocentesis is **legal** for detecting genetic disorders, but **ILLEGAL** for sex determination in India under PCPNDT Act.

Q5

Describe two principal procedures adopted for test tube baby / ART. Explain ZIFT and how IUT differs from it.

Answer:

1. ZIFT (Zygote Intra Fallopian Transfer):

- Ova collected from wife + semen from husband → fertilised *in vitro* in laboratory.
- Resulting **zygote or early embryo** (up to 8-cell stage) is transferred into the **fallopian tube**.
- Also called IVF-ET when embryo is transferred to uterus.

2. IUT (Intra Uterine Transfer):

- Similar to ZIFT, but the embryo (at **8-celled stage or beyond**) is transferred into the **uterus** (not fallopian tube).

Difference between ZIFT and IUT:

ZIFT	IUT
Embryo transferred to **fallopian tube**	Embryo transferred to **uterus**
Embryo at **zygote/early stage**	Embryo at **8-cell stage or later**

⋆ Key Point

Other ART: GIFT (Gamete Intra Fallopian Transfer), ICSI (Intra Cytoplasmic Sperm Injection), AI (Artificial Insemination).

Chapter 5 — Principles of Inheritance and Variation

Q1

Name the pattern of inheritance where F1 phenotype (i) resembles only one parent (ii) does not resemble either parent and is in between.

Answer:

- **(i) F1 resembles only one of the two parents:** **Complete Dominance** (Mendelian Dominance)
  Example: Tall × Dwarf → All Tall in F1 (Tallness is dominant over Dwarfness).
- **(ii) F1 does not resemble either parent (intermediate):** **Incomplete Dominance**
  Example: Red flower (RR) × White flower (rr) → Pink flower (Rr) in F1 (snapdragon/antirrhinum).
  F2 ratio: **1 Red : 2 Pink : 1 White** (phenotypic = genotypic ratio, 1:2:1).

⋆ **Key Point**

Incomplete dominance F2 = 1:2:1 phenotypic ratio (same as genotypic ratio). Original phenotypes reappear ∴ not blending.

Q4

Why did TH Morgan select *Drosophila melanogaster* to study sex-linked genes?

Answer:

Reasons for choosing *Drosophila melanogaster*:

1. **Short life cycle** (≈2 weeks) → many generations can be studied quickly.
2. **Large number of progeny** (hundreds per mating) → statistically significant data.
3. **Small number of chromosomes** (2n = 8, only 4 pairs) → linkage studies easy.
4. **Clear sexual dimorphism** → males and females easily distinguishable.
5. **Easy and inexpensive to maintain** in lab on fermenting fruit/banana.
6. **Many mutant strains** readily available (white eye, yellow body, vestigial wings, etc.).

⋆ **Key Point**

Scientific name: *Drosophila melanogaster* — Common name: Fruit fly — 2n = 8 chromosomes (3 pairs autosomes + 1 sex chromosome pair).

Q6

In snapdragon, RR × rr showed all pink flowers in F1. (i) Why is the appearance of pink flowers not called blending? (ii) What is this phenomenon called?

Answer:

(i) Why it is NOT blending:

- If it were true blending, the red and white traits would permanently mix and could *never* be recovered separately.
- However, when F1 pink flowers (Rr) are self-crossed: F2 gives **1 Red (RR) : 2 Pink (Rr) : 1 White (rr)**.
- The original red and white phenotypes **reappear in F2** — proving genes maintain their identity and have NOT blended.

(ii) This phenomenon is called: Incomplete Dominance

- Neither allele (R or r) is completely dominant over the other.
- The heterozygote (Rr) shows an **intermediate (pink) phenotype**.
- F2 phenotypic ratio = **1:2:1** (same as genotypic ratio — unlike Mendel's 3:1).

Q9

Explain with an example the inheritance where two different dominant alleles express themselves simultaneously. Name this pattern.

Answer:

This pattern is called: Codominance

Both alleles are fully and simultaneously expressed in the heterozygote — neither masks the other.

Example: ABO Blood Group System (I^A and I^B alleles)

- I^A allele codes for enzyme producing **A antigen** on RBCs.
- I^B allele codes for enzyme producing **B antigen** on RBCs.
- When genotype is I^AI^B → **Blood Group AB** is expressed.
- Both A and B antigens are present on RBC surface *simultaneously*.
- Neither I^A nor I^B is dominant over the other — *both are fully expressed*.

⋆ Key Point

Codominance ≠ Incomplete dominance. In codominance, BOTH traits show FULLY. In incomplete dominance, an INTERMEDIATE trait appears. ABO blood group also shows **Multiple allelism** (3 alleles: I^A, I^B, i).

Q10

State and explain the Law of Independent Assortment using a Mendelian dihybrid cross.

Answer:

Law of Independent Assortment (Mendel's Second Law):

When two pairs of contrasting characters are considered simultaneously, the alleles of one pair **segregate independently** of the alleles of the other pair during gamete formation.

Dihybrid Cross — Seed Shape × Seed Colour in Pea:

- P: **RRYY** (Round Yellow) × **rryy** (Wrinkled Green)

- F1: All **RrYy** (Round Yellow) — both dominant traits appear
- F1 × F1: RrYy × RrYy
- Gametes produced by each parent: **RY, Ry, rY, ry** (4 types each)

F2 Phenotypic Ratio:

$$\underbrace{\text{9 Round Yellow}}_{\text{R_Y_}} : \underbrace{\text{3 Round Green}}_{\text{R_yy}} : \underbrace{\text{3 Wrinkled Yellow}}_{\text{rrY_}} : \underbrace{\text{1 Wrinkled Green}}_{\text{rryy}}$$

This **9:3:3:1 ratio** proves that genes for seed shape and seed colour assort *independently* into gametes.

⋆ Key Point

9:3:3:1 is the hallmark dihybrid ratio proving Independent Assortment. Holds true only for genes on *different* (non-homologous) chromosomes.

Chapter 6 — Molecular Basis of Inheritance

Q1

Why is RNA more reactive in comparison to DNA?

Answer:

RNA is more reactive than DNA due to the following structural differences:

1. RNA has **ribose sugar** with a **free –OH group at the 2' carbon** → makes RNA chemically reactive and susceptible to alkaline hydrolysis.
2. DNA has **deoxyribose sugar** (lacking –OH at 2' carbon) → chemically stable.
3. RNA contains **Uracil** (instead of Thymine) → Uracil lacks the methyl group, making RNA less thermally stable.
4. RNA is generally **single-stranded** → more exposed to enzymatic (RNase) degradation.
5. DNA is **double-stranded** → hydrogen bonds between complementary strands confer chemical stability.

⋆ **Key Point**

DNA = stable genetic material (store info). RNA = reactive functional molecule (expresses info). 2'-OH in ribose = key chemical difference.

Q2

Name the enzyme and state its property responsible for continuous and discontinuous replication of the two strands of a DNA molecule.

Answer:

Enzyme: DNA-dependent DNA Polymerase III

Key Property: DNA polymerase can synthesise DNA **only in the 5'→3' direction** by adding nucleotides to the 3'-OH end of the growing strand. It *cannot* synthesise in the 3'→5' direction.

Since both strands of DNA are antiparallel:

- **Leading strand (continuous):** Template runs 3'→5', so new strand is synthesised continuously in 5'→3' direction as the replication fork opens.
- **Lagging strand (discontinuous):** Template runs 5'→3', so new strand cannot be synthesised continuously. Instead, short fragments called **Okazaki fragments** are synthesised (each in 5'→3'), later joined by **DNA Ligase**.

⋆ **Key Point**

DNA Pol III = main replication enzyme | DNA Ligase = joins Okazaki fragments | Primase = lays RNA primer | Helicase = unwinds DNA

Q4

Describe the structure of a nucleosome. How do histones acquire a positive charge?

Answer:

Nucleosome — Structure:
A nucleosome is the basic repeating structural unit of chromatin in eukaryotes.

- **Core:** 8 histone proteins (octamer) = 2 each of **H2A, H2B, H3, H4**.
- **DNA wrapping:** ≈200 bp of DNA wraps around this histone octamer (≈1.65 turns).
- **Linker DNA:** ≈80 bp of DNA connects adjacent nucleosomes (the "string" in "beads on a string").
- **H1 histone (linker histone):** Binds at the entry/exit point of DNA; involved in higher-order chromatin packaging.

How histones acquire positive charge:
Histones are rich in basic amino acids — **lysine** and **arginine** — which carry positive charges at physiological pH. This positive charge enables electrostatic interaction with the **negatively charged** DNA backbone (phosphate groups).

⋆ Key Point

"Beads on a string" = nucleosomes under electron microscope. 1 nucleosome ≈ 200 bp DNA + histone octamer. Nucleosomes are found in eukaryotic chromatin.

Q5

Differentiate between exons and introns.

Answer:

Exons	Introns
Coding sequences of a gene	Non-coding intervening sequences
Present in mature mRNA after splicing	Removed from hnRNA during splicing
Expressed and translated into protein	Not translated into protein
Also called "expressed sequences"	Also called "intervening sequences"
Found in both prokaryotes and eukaryotes	Found mainly in eukaryotes

Q6

Describe the structure and function of a tRNA molecule. Why is it referred to as an adapter molecule? Explain splicing of hnRNA.

Answer:

Structure of tRNA (Cloverleaf secondary structure):

- Single-stranded RNA that folds into a **cloverleaf** due to intramolecular base pairing.
- **4 arms/loops:**
  1. **Acceptor stem (3' end):** Has CCA-OH sequence; binds to specific amino acid (charged by aminoacyl-tRNA synthetase).
  2. **Anticodon loop:** Contains the **anticodon** (3 nucleotides complementary to mRNA

codon).

3. **D-loop (Dihydrouracil loop):** Site for recognition by aminoacyl-tRNA synthetase.
4. **TψC loop:** Binds to ribosome.

- Actual 3D structure: **L-shaped (inverted-L)** molecule.

Why called Adapter Molecule:
tRNA acts as an adapter between the **nucleotide language** of mRNA (codons) and the **amino acid language** of proteins:

- Anticodon loop reads the mRNA codon
- Acceptor stem carries the specific amino acid

Splicing of hnRNA:
The primary RNA transcript (hnRNA) contains both introns and exons. Introns are removed (spliced out) by a complex called the **spliceosome**. The remaining exons are joined together to form mature mRNA, which exits the nucleus for translation.

⋆ Key Point

tRNA = adapter between nucleotide (mRNA) language and amino acid (protein) language. 1 tRNA per amino acid (20 types of tRNA for 20 amino acids).

Q7

Sketch a schematic diagram of the lac operon in switched-on position. How is the operon switched off?

Answer:

Lac Operon — Components:

- **Regulator gene (i):** Produces repressor protein (always).
- **Promoter (P):** Binding site for RNA polymerase.
- **Operator (O):** Repressor binding site (between promoter and structural genes).
- **Structural genes:** *lacZ* (β-galactosidase), *lacY* (permease), *lacA* (transacetylase).

SWITCHED ON (Inducer/lactose present):

- Lactose (inducer) binds to repressor protein → changes its conformation → repressor **CANNOT bind** to operator.
- RNA polymerase binds to promoter freely → transcribes structural genes → enzymes produced to metabolise lactose.

SWITCHED OFF (No lactose):

- Repressor protein is **active** and **binds to the operator**.
- RNA polymerase **cannot proceed** past the operator → structural genes are NOT transcribed → no enzyme production.

Operon diagram: `[i]---[P]---[O]---[lacZ]---[lacY]---[lacA]`

⋆ Key Point

Lac operon = **inducible operon**. Inducer = allolactose (metabolite of lactose). Discovered by **Jacob and Monod** (1961). Negative regulation by repressor.

Chapter 7 — Evolution

Q1

Name the type of evolution that resulted in development of wings of butterfly and birds. What are such structures called?

Answer:

Type of Evolution: Convergent Evolution
Different (unrelated) organisms evolve similar structures independently due to similar environmental pressures (natural selection).
Such structures are called: Analogous Structures

- Wings of butterfly = membranous extensions of insect body wall.
- Wings of birds = modified forelimbs (pentadactyl limb of vertebrates).
- Same *function* (flight) but **different evolutionary origins**.

Contrast:

- **Divergent Evolution → Homologous structures** (same origin, different functions).
- Example: Forelimbs of whale, bat, cheetah, human — all modified pentadactyl limbs.

⋆ Key Point

Analogy = similar function, different origin (convergent). **Homology** = same origin, different function (divergent).

Q4

How does the Hardy-Weinberg equation explain genetic equilibrium? Describe how this equilibrium gets disturbed, leading to the founder effect.

Answer:

Hardy-Weinberg Principle:
Allele frequencies in a population remain constant from generation to generation *if no evolutionary forces are acting.*
Mathematical Expression:

$$p^2 + 2pq + q^2 = 1$$

Where: p = frequency of dominant allele (A); q = frequency of recessive allele (a)
p^2 = freq of AA; $2pq$ = freq of Aa; q^2 = freq of aa; and $p + q = 1$.
Conditions for equilibrium: Large population, random mating, no mutation, no natural selection, no gene flow, no genetic drift.
Factors disturbing H-W equilibrium (5 forces of evolution):

1. Gene migration (gene flow) 2. Genetic drift 3. Mutation
2. Natural selection 5. Non-random mating

Founder Effect:

- When a small group of individuals (founders) separates from a larger population and establishes a new colony.

- The new population has very limited genetic diversity (only founders' alleles).
- Allele frequencies in new population differ drastically from original by *chance*, not selection.
- This is a form of **genetic drift**.
- **Example:** Ellis-van Creveld syndrome in the Amish population of USA.

⋆ Key Point

Founder effect = type of genetic drift. Small founding population → reduced genetic variation → chance fixation of some alleles.

Q7

Explain the phenomenon of evolution by natural selection as supported by white-winged and dark-winged moth populations in England (1850–1920).

Answer:

Industrial Melanism in *Biston betularia* (Peppered Moth):
Before 1850 (Pre-industrial period):

- Tree barks were light-coloured (covered with light lichen).
- **White/peppered moths** were well camouflaged → survived predation (selected FOR).
- **Dark (melanistic) moths** were visible on light bark → easily eaten by birds (selected AGAINST).
- Population: predominantly white moths.

After 1850 (Industrial Revolution):

- Industrial pollution killed lichens; soot deposited → tree bark became dark.
- White moths became VISIBLE on dark bark → easy prey for birds (selected AGAINST).
- Dark moths were now CAMOUFLAGED → survived and reproduced (selected FOR).
- By 1920: Population shifted to predominantly dark moths.

Conclusion: This demonstrates **Natural Selection in action**. The environment "selected" the better-adapted variant.

⋆ Key Point

Industrial melanism = classic example of **directional selection**. Population mean shifted toward one extreme (dark colouration). After clean air acts, peppered moths regained dominance.

Q9

Differentiate between homology and analogy. Give one example of each.

Answer:

Homology	Analogy
Structures with same origin (common ancestry)	Structures with different evolutionary origin
May perform different functions	Perform same/similar functions
Evidence of **divergent evolution**	Evidence of **convergent evolution**
Example: Forelimbs of whale, bat, human, horse — all modified pentadactyl limbs with same bone structure	Example: Wings of insects and birds — different structure, same function (flight)
Example: Thorns of Bougainvillea vs. tendrils of Cucurbita (both leaf modifications)	Example: Sweet potato (root modification) vs. potato (stem modification) — similar appearance

Chapter 8 — Human Health and Diseases

Q5

Name and explain the type of immunity provided by colostrum and vaccinations. Name the antibodies present.

Answer:

(i) Colostrum → Passive Immunity:

- Newborn receives **pre-formed antibodies** directly from mother through colostrum (first milk produced after birth).
- Baby's own immune system does *not* produce these antibodies.
- Called *passive* because the infant is a passive recipient.
- Provides **immediate but short-term** protection (no memory cells formed).
- **Antibody in colostrum: IgA** (Immunoglobulin A).

Vaccination → Active Immunity:

- A weakened/killed pathogen or its antigen is introduced into the body.
- The person's own immune system recognises it → produces antibodies and **memory cells** (B and T memory cells).
- **Long-lasting protection** — immune system is "activated" and can respond rapidly to future infection.

(ii) Antibody produced in response to allergens: IgE (Immunoglobulin E).

⋆ Key Point

Colostrum = IgA (passive, immediate, short-lived) | Allergy = IgE | Vaccination = Active immunity (IgG mainly, long-lasting)

Q7

Name the parasite that causes filariasis. Mention its two diagnostic symptoms. How is it transmitted?

Answer:

Causative Parasite: *Wuchereria bancrofti* (also *W. malayi*)
Commonly called the filarial worm (a parasitic nematode).

Diagnostic Symptoms:

1. **Lymphoedema** — chronic inflammation and swelling of lymph nodes and lymphatic vessels due to the worm's presence.
2. **Elephantiasis** — extreme swelling of lower limbs (legs), scrotum, or other body parts due to accumulation of lymph fluid; skin becomes thick and rough.

Mode of Transmission:

- Transmitted through the bite of the **female *Culex* mosquito** (the vector).
- Infected mosquito bites human → filarial larvae (microfilariae) enter blood → migrate

to lymphatic system → cause damage over years.

⋆ Key Point

Filariasis: Parasite = *Wuchereria bancrofti* | Vector = *Culex* mosquito | Symptom = Elephantiasis (lymphatic obstruction)

Q8

Differentiate between active and passive immunity.

Answer:

Active Immunity	Passive Immunity
Body produces its own antibodies in response to antigen	Pre-formed antibodies are directly introduced
Slow response — takes time to develop	Immediate protection
Long-lasting (memory cells formed)	**Short-lived** (no memory cells)
Obtained by: natural infection or vaccination	Obtained by: colostrum, anti-sera, anti-venom injection
Example: Smallpox vaccine, polio vaccine	Example: Tetanus anti-toxin, snake anti-venom, colostrum

Q9

State the function of interferons.

Answer:

Interferons — Functions:

- Interferons are **glycoproteins** produced by **virus-infected cells**.
- Released by infected cells; act on **neighbouring uninfected cells** as a warning signal.
- Stimulate uninfected cells to produce **antiviral proteins** → prevents viral replication in those cells.
- Interferons do *not* kill the virus directly — they make surrounding cells **resistant** to viral infection.
- Also activate **NK cells (Natural Killer cells)** and macrophages to attack virus-infected cells.
- Example: **Interferon-alpha** is used therapeutically against hepatitis and certain cancers.

⋆ Key Point

Interferons = early warning system. Infected cell → produces interferon → neighbouring cells become resistant. Named after their ability to "interfere" with viral replication.

Chapter 10 — Microbes in Human Welfare

Q2

How does the activity of (i) Mycorrhiza (ii) Cyanobacteria (iii) Rhizobium help in organic farming?

Answer:

(i) Mycorrhiza:

- Symbiotic association between **fungi** and **roots** of higher plants.
- Fungal hyphae greatly increase surface area for water and mineral absorption from soil.
- Particularly important for **phosphorus absorption** from soil.
- Provides resistance to root-borne pathogens, drought, and salinity stress.
- Promotes plant growth without chemical fertilisers.

(ii) Cyanobacteria (Blue-Green Algae):

- Autotrophic microbes that perform **biological nitrogen fixation**.
- Examples: *Anabaena, Nostoc, Oscillatoria* — fix atmospheric N_2 → add nitrogen to soil.
- Important biofertiliser in **paddy/rice fields** (paddy fields remain flooded — ideal for cyanobacteria).
- Also add organic matter to soil, improving fertility.

(iii) Rhizobium:

- Symbiotic nitrogen-fixing bacteria found in **root nodules of leguminous plants** (peas, beans, groundnut).
- Convert atmospheric N_2 → ammonia (NH_3) through biological nitrogen fixation.
- Provide fixed nitrogen to the host plant; enrich soil after crop decomposition.

⋆ **Key Point**

Rhizobium = symbiotic | Azotobacter, Azospirillum = free-living N_2 fixers | Anabaena = N_2 fixer in rice fields | Mycorrhiza = phosphorus absorption

Q5

How can sewage be used to generate biogas? Explain.

Answer:

1. **Primary Treatment:** Sewage is filtered and sedimented to remove large solids.
2. **Secondary Treatment (Aerobic):** Effluent is aerated in large tanks; aerobic microbes break down organic matter → reduces BOD (Biological Oxygen Demand).
3. **Sludge Formation:** Bacteria + settling organic matter forms **activated sludge**.
4. **Anaerobic Digestion:** A portion of sludge is pumped into large **anaerobic sludge digesters**.

- Anaerobic bacteria — especially **methanogens** (e.g., *Methanobacterium*) — decompose organic matter.
- Produce: **CH_4 (methane) + CO_2 + H_2S = Biogas** (≈65% methane).

5. **Biogas Collection:** Collected and used as fuel for cooking, electricity generation, or heating.

The treated effluent (after secondary treatment) is released into rivers or ponds.

⋆ **Key Point**

BOD = Biological Oxygen Demand (higher BOD = more polluted). Biogas ≈ 65% methane. Methanogen = key microbe in biogas production.

Chapter 11 — Biotechnology: Principles and Processes

Q6

Explain the significance of palindromic nucleotide sequences in formation of recombinant DNA. Write the use of restriction endonuclease.

Answer:

(i) Significance of Palindromic Sequences:

- Palindromic sequences read the same on both strands of DNA in the 5'→3' direction.
- Example: `5'--GAATTC--3'` / `3'--CTTAAG--5'`
- Restriction endonucleases recognise these specific palindromic sequences and cut DNA.
- Cutting at palindromic sites produces **sticky ends** (single-stranded overhangs) or blunt ends.
- Sticky ends allow foreign DNA to be inserted easily — they base-pair with complementary sticky ends of the vector.
- This is fundamental to creating **recombinant DNA**.

(ii) Use of Restriction Endonuclease:

- Acts as **"molecular scissors"** — cuts DNA at specific palindromic sequences.
- Produces fragments with sticky or blunt ends for cloning.
- Used in RFLP analysis, gene mapping, and making recombinant vectors.
- Allows isolation of specific genes of interest.

⋆ Key Point

EcoRI: recognises `5'-GAATTC-3'`, cuts between G and A → produces 4-nucleotide 5' sticky ends (AATT overhangs).

Q9

Explain the mode of action of EcoRI. How does restriction nuclease differ from an exonuclease?

Answer:

EcoRI — Mode of Action:

- EcoRI is a restriction endonuclease from *Escherichia coli* (strain R, first enzyme identified).
- Recognises the palindromic sequence: `5'--G↓AATTC--3'` / `3'--CTTAA↑G--5'`
- Cuts between **G and A** on both strands at staggered positions.
- Produces **sticky ends (cohesive ends)** — 4-nucleotide 5' overhangs (`AATT`).
- Sticky ends can base-pair with complementary sticky ends of other DNA fragments cut by EcoRI.
- These annealed pieces are permanently joined by **DNA Ligase**.

Restriction Endonuclease vs. Exonuclease:

Restriction Endonuclease	Exonuc
Cuts DNA at **internal specific sites**	Remove 3')
Sequence-specific (palindromic recognition)	No sequ
Produces defined fragments	Degrade

⋆ Key Point

EcoRI = endonuclease (cuts internally at GAATTC). Exonuclease = cuts from DNA ends. Restriction enzymes are named after the organism: **E**scherichia **co**li strain **R** → EcoRI.

Q10

Why do DNA fragments move towards the anode during gel electrophoresis?

Answer:

- DNA is a **negatively charged molecule** — due to the **phosphate groups** in its backbone (each phosphate carries a negative charge at physiological pH).
- During gel electrophoresis, an **electric field** is applied across the agarose gel.
- Negatively charged DNA fragments are **attracted towards the positive electrode (anode)**.
- They migrate through the agarose gel from cathode (–) to anode (+).
- **Smaller fragments** experience less resistance through gel pores → travel *faster and farther*.
- **Larger fragments** move slower → remain closer to the wells.
- Bands visualised using ethidium bromide staining under UV light.

⋆ Key Point

DNA → negatively charged (phosphate groups) → moves toward anode (+). Smaller fragments = farther migration = lower molecular weight.

Chapter 12 — Biotechnology and its Applications

Q3

Name the source of insulin extraction earlier. Explain the process of synthesis of insulin by Eli Lilly company. Name the technique used.

Answer:

(i) Earlier Source of Insulin:
Insulin was extracted from the **pancreas of slaughtered cattle and pigs**. Problem: caused allergic reactions in some patients (slight structural difference from human insulin).
(ii) Synthesis of Insulin by Eli Lilly Company (Recombinant DNA Technology):
Human insulin (humulin) consists of two chains: **Chain A** (21 amino acids) + **Chain B** (30 amino acids) connected by **disulfide bonds**.

1. DNA sequences encoding Chain A and Chain B were **synthesised artificially** (in vitro).
2. Each sequence was inserted into **separate plasmid vectors** of *E. coli*.
3. *E. coli* expressed each chain separately as **fusion proteins** with β-galactosidase.
4. Chains were extracted and **purified**.
5. Chain A and Chain B were **combined by forming disulfide bonds in vitro** $\rightarrow$ functional human insulin (humulin).

Technique used: Recombinant DNA Technology (rDNA technology)
(iii) Difference: Recombinant insulin (humulin) is *identical* to human insulin; animal-extracted insulin had slight structural differences causing immune reactions.

⋆ **Key Point**

Humulin = first commercial recombinant protein (1982, Eli Lilly). Two chains A + B joined by disulfide bonds. rDNA technology allows human-sequence insulin production.

Q6

How does RNA interference take place in eukaryotes? Mention its importance.

Answer:

RNA Interference (RNAi) — Process:
RNAi is a cellular mechanism of **gene silencing** using double-stranded RNA (dsRNA).

1. dsRNA (double-stranded RNA) is introduced into or produced within the cell.
2. An enzyme called **Dicer** cleaves the dsRNA into small fragments of 21–23 bp called **siRNA (small interfering RNA)**.
3. siRNA is incorporated into a protein complex called **RISC (RNA-Induced Silencing Complex)**.
4. The siRNA strands separate; one strand (guide strand) remains in RISC and guides it to **complementary mRNA**.
5. RISC cleaves/degrades the target mRNA $\rightarrow$ gene is silenced (not translated into protein).

Importance / Applications:

- Used to protect plants from pests — tobacco plant protected from nematode *Meloidogyne incognita* using RNAi.
- Potential therapeutic tool against viral infections and cancer.
- Powerful research tool to study gene function by silencing specific genes.

⋆ **Key Point**

RNAi = gene silencing using dsRNA → siRNA → RISC → mRNA degradation. Key application: protecting tobacco from *Meloidogyne incognita* nematode.

Chapter 13 — Organisms and Populations

Q1

Explain with the help of a graph the population growth curve when resources are (A) limiting (B) not limiting. Explain carrying capacity.

Answer:

(A) When resources are NOT limiting — Exponential (J-shaped) Growth:

- Population grows without any limiting factors (unlimited food, space).
- Formula: $\frac{dN}{dt} = rN$ (r = intrinsic rate of natural increase)
- Graph: **J-shaped curve** — population keeps increasing indefinitely.
- Rarely found in nature; may occur when a species enters a new habitat.

(B) When resources ARE limiting — Logistic (S-shaped/sigmoid) Growth:

- Resources (food, space) are limited → intraspecific competition occurs.
- Growth rate slows as population size (N) approaches carrying capacity (K).
- Formula: $\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$
- Graph: **S-shaped (sigmoid) curve** — growth initially accelerates, then decelerates, and eventually stabilises.
- This is more realistic in nature.

Carrying Capacity (K):

- Maximum number of individuals of a species that a habitat can sustain with available resources.
- When $N = K$, growth rate = 0 (population stabilises).
- Also called the **saturation density**.

⋆ **Key Point**

J-curve = unlimited resources = exponential growth. S-curve = limited resources = logistic growth. K = carrying capacity. $(K-N)/K$ = feedback factor.

Q5

Describe the responses of animals to abiotic factors: Regulate, Conform, Migrate, Suspend. Calculate death rate if 8 individuals in a population of 80 butterflies die in a week.

Answer:

Responses to abiotic factors:

- **(a) Regulate:** Organism maintains constant internal environment (homeostasis) despite external changes.
  Example: Humans, birds, mammals — maintain constant body temperature (thermoregulation). Called **endotherms/homeotherms**.

- **(b) Conform:** Cannot regulate internal conditions — internal environment changes with external environment.
  Example: Most invertebrates, fishes — body temperature varies with surroundings (**poikilotherms/ectotherms**).
- **(c) Migrate:** Move temporarily to more hospitable environments to escape stressful conditions.
  Example: Siberian cranes migrate from Siberia to Bharatpur/Chilika Lake during winter.
- **(d) Suspend:** Suspend metabolic activities during stressful periods (dormancy).
  - **Hibernation** = winter sleep (bear, bat)
  - **Aestivation** = summer dormancy (snails, lungfish)
  - **Diapause** = suspended development in zooplankton under unfavourable conditions

Death Rate Calculation:

$$\text{Death rate} = \frac{\text{Number of deaths}}{\text{Total population}} = \frac{8}{80} = 0.1 \text{ (or 10\% per week)}$$

⋆ **Key Point**

Regulate = homeostasis | Conform = body changes with environment | Migrate = escape | Suspend = dormancy (hibernate/aestivate/diapause)

Chapter 14 — Ecosystem

Q1

How does a detritivore differ from a decomposer? Explain with an example of each.

Answer:

Detritivore	Decomposer
Physically breaks down dead organic matter (detritus) into smaller pieces	Chemically breaks down organic matter into inorganic substances
Mechanical fragmentation — increases surface area for decomposers	Enzymatic digestion (secretes extracellular enzymes)
Macroscopic organisms	Microscopic organisms (bacteria and fungi)
Examples: Earthworm, millipede, woodlice, dung beetle	Examples: *Pseudomonas* (bacteria), *Rhizopus, Aspergillus* (fungi)
Produces smaller detritus particles for decomposers to act on	Produces final inorganic products: CO_2, H_2O, minerals

Q2

Write the equation that helps in deriving the net primary productivity of an ecosystem. How is it derived?

Answer:

Equations:

$$\text{NPP} = \text{GPP} - R$$

Where:

- **GPP** = Gross Primary Productivity (total energy fixed by autotrophs via photosynthesis)
- **R** = Respiration losses by producers
- **NPP** = Net Primary Productivity (energy available for consumers; biomass available for herbivores)

$$\text{Secondary Productivity} = \text{Energy stored by consumers using NPP}$$

NPP is expressed in g/m^2/yr (dry matter) or kcal/m^2/yr.

⋆ **Key Point**

GPP − Respiration = NPP. NPP is the energy available for the next trophic level. **Lindeman's 10% law**: only 10% of energy is transferred from one trophic level to the next.

Q5

Explain the carbon cycle with the help of a simplified model.

Answer:

Carbon exists in atmosphere as CO_2 (≈0.04%). Key processes:

1. **Photosynthesis (CO_2 fixation):**
   Plants/algae absorb CO_2 from atmosphere → convert to organic compounds (glucose) using sunlight + chlorophyll.
2. **Consumption:**
   Herbivores eat plants → carbon transfers; carnivores eat herbivores → further transfer up food chain.
3. **Respiration (CO_2 release):**
   All organisms release CO_2 back to atmosphere through cellular respiration (aerobic breakdown of organic molecules).
4. **Decomposition:**
   Decomposers break down dead organic matter → release CO_2 and minerals to soil and atmosphere.
5. **Combustion:**
   Burning of fossil fuels (coal, petroleum) and wood releases stored carbon as CO_2 → main cause of elevated atmospheric CO_2 (greenhouse effect, global warming).
6. **Sedimentation:**
   Carbon from marine organisms forms calcium carbonate shells/skeletons → deposited as sedimentary rocks (limestone) over geological time.

Simplified model: Atmosphere ↔ Producers (photosynthesis/respiration) ↔ Consumers ↔ Decomposers ↔ Soil ↔ Fossil fuels (combustion) → Atmosphere

⋆ **Key Point**

Carbon cycle integrates: atmosphere ↔ living organisms ↔ soil ↔ ocean ↔ fossil fuels. Combustion of fossil fuels = main anthropogenic disruption.

Q6

How does the phosphorus cycle differ from the carbon cycle?

Answer:

Carbon Cycle	Phosphorus Cycle
Carbon circulates through atmosphere, biosphere, hydrosphere	Phosphorus has **no atmospheric phase** — mainly lithosphere & hydrosphere
CO_2 is the main reservoir (atmosphere)	Rocks and soil are the main reservoir
Rapid cycling (photosynthesis, respiration)	Slow cycling (geological processes — weathering, erosion)
Involves gaseous state (CO_2, CH_4)	Does not involve gaseous compounds
Easily replenished via atmospheric exchange	Replenishment is very slow (depends on rock weathering)

⋆ Key Point

Carbon cycle = **gaseous cycle** (atmosphere involved). Phosphorus cycle = **sedimentary cycle** (no atmospheric phase; rocks are reservoir).

Chapter 15 — Biodiversity and Conservation

Q1

Why should we conserve biodiversity? How can we do it? Explain the importance of biodiversity hotspots and sacred groves.

Answer:

Why conserve biodiversity?

- **Narrowly utilitarian:** Direct economic benefits — food, medicine (25% of drugs from plants), timber, fibre, industrial products.
- **Broadly utilitarian:** Ecosystem services — oxygen production, pollination, water purification, climate regulation, nutrient cycling, soil formation.
- **Ethical:** Every species has intrinsic value and the right to exist irrespective of its utility to humans.

How to conserve:

- **In situ conservation (on-site):** Protecting species in their natural habitat.
  Examples: National parks, wildlife sanctuaries, biosphere reserves, sacred groves.
- **Ex situ conservation (off-site):** Protecting outside natural habitat.
  Examples: Zoological parks, botanical gardens, seed banks, cryopreservation, tissue culture.

Biodiversity Hotspots:

- Areas with **exceptionally high species richness + high endemism + severe threat of habitat loss**.
- 34 hotspots identified globally.
- India's hotspots: Western Ghats + Sri Lanka, Eastern Himalayas, Indo-Burma, Sundaland.
- Receive priority conservation funding and efforts.

Sacred Groves:

- Forest tracts protected by tribal/local communities based on religious and cultural beliefs.
- No human disturbance allowed; serve as **refugia** for rare and threatened species.
- Maintain local biodiversity without formal legal protection.
- Examples: Khasi & Jaintia hills (Meghalaya), Aravalli hills, Western Ghats.

⋆ **Key Point**

Sacred groves = community-protected forests. Hotspots = high biodiversity + high endemism + high threat. India = megadiversity country (12 of 34 hotspots nearby).

Q3

Write the importance of cryopreservation in conservation of biodiversity.

Answer:

Cryopreservation — Importance:

- Cryopreservation involves storing biological material (gametes, embryos, seeds, tissues) at very low temperatures (usually in **liquid nitrogen at −196°C**).
- **Long-term storage** of genetic material without degradation or loss of viability.
- Useful for conserving species/varieties that are threatened with extinction.
- **Seed banks** use cryopreservation to store seeds of crop wild relatives and rare plants.
- Allows conservation of genetic diversity of domesticated animals through semen and embryo banks.
- Can be used later to restore populations or breed endangered species (assisted reproduction).
- Examples: National Bureau of Plant Genetic Resources (NBPGR), New Delhi — stores seeds cryogenically.

⋆ Key Point

Cryopreservation = ex situ conservation method. Liquid nitrogen at −196°C. Used in seed banks, gene banks, sperm/embryo banks.

Q4

Write the importance of species diversity to the ecosystem. Support your answer with Tilman's finding.

Answer:

Importance of Species Diversity:

- Greater species diversity → greater **ecosystem stability** and productivity.
- Diverse ecosystems are more **resilient to disturbances** (natural or anthropogenic).
- Diverse food webs have multiple energy pathways — less vulnerable to loss of a single species.
- More species = greater efficiency in resource utilisation (niche partitioning).
- Each species performs unique ecosystem functions — loss leads to functional impairment.

Tilman's Finding (long-term grassland experiments):

- David Tilman conducted long-term biodiversity experiments on grassland plots.
- Plots with **more species** showed **less year-to-year variation in total biomass** (more stable, more consistent productivity).
- Plots with higher diversity showed **higher overall productivity**.
- Loss of diversity led to reduced ecosystem functioning.

⋆ Key Point

Tilman: Higher biodiversity = higher stability + higher productivity in grassland ecosystems. "Rivet hypothesis" of Paul Ehrlich: each species is like a rivet in an aeroplane — losing too many causes collapse.

Q5

Name the type of biodiversity represented by: (i) 1000 varieties of mangoes in India (ii) Variations in reserpine in *Rauwolfia vomitoria* from different Himalayan regions.

Answer:

- **(i) 1000 varieties of mangoes (*Mangifera indica*) in India:**
  This represents **Genetic Diversity** — variations within the *same species* at the genetic level.
  India is a "mega-diversity" country partly because of this vast intraspecific diversity.
- **(ii) Variations in potency and concentration of reserpine in *Rauwolfia vomitoria* from different regions of Himalayas:**
  This also represents **Genetic Diversity** — the same species shows different genotypes due to different environmental/geographic pressures (ecotypes).

Genetic Diversity	Species Diversity	Ecological Diversity
Variation within a species	Variation between species	Variation in ecosystems
1000 mango varieties; Rauwolfia potency variation	Number of species in an area	Forests, wetlands, deserts, grasslands

⋆ **Key Point**

Three levels of biodiversity: **Genetic** (within species) | **Species** (between species) | **Ecological** (ecosystem level). All three levels need conservation.

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Refer to NCERT textbook diagrams for all labelled drawings.
Practice writing answers in CBSE format within the word limit.